CMSC/DATA 320 – Fall 2025 (Section 0201)

Description Schedule Deliverables

Introduction to Data Science

DATA SCIENCE
SCIGIA

Instructor:

Dr. Fardina Alam
(fardina@umd.edu)

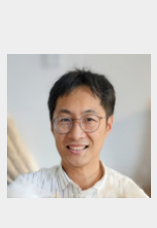
Head Teaching Assistants (TAs):

Aviva Prins

Guanhong Wang

Teaching Assistants (TAs):

Yonghan Lee

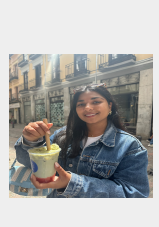


Shengjie Xu

Weize Liu

Fnu Sakshi

Taewon Kang



Tanya Gupta

Lecture Schedule: IRB 0324: 3:30 PM - 4:45 PM EDT, Tuesday & Thursday.

Course Overview

Data science encapsulates the interdisciplinary activities required to create data-centric products and applications that address specific scientific, socio-political, or business questions. It has drawn tremendous attention from both academia and industry and is making deep inroads in industry, government, health, and journalism.

This course focuses on (i) data management systems, (i) exploratory and statistical data analysis, (ii) data and information visualization, and (iv) the presentation and communication of analysis results. It will be centered around case studies drawing extensively from applications, and will yield a publicly-available final project or tutorial that will strengthen course participants' data science portfolios.

The course will primarily consist of sets of self-contained lectures and assignments that leverage real-world data science platforms when needed. As such, there is no assigned textbook, but there will be some recommended ones. Many lectures will come with links to required readings, which should be completed before or after the lecture (as declared), and, when appropriate, a list of links to other web resources.

Note: This course outline is tentative and subject to modification to meet the specific needs and requirements of the students and the evolving field of data science.

Prerequisites

Students enrolled in the course should be comfortable with programming (for those at UMD, passing CMSC216 will be sufficient) and have a reasonable level of mathematical maturity. The course will primarily use the
Python
programming language through
Google Colab
(or
Jupyter
Notebook if needed), with support from the
Anaconda
package manager. Primer lectures on Python for data science will be provided early in the semester, so there is no need to worry if you are new to Python. Later lectures will introduce statistics and machine learning, potentially incorporating basic calculus and linear algebra. A light level of mathematical maturity, roughly equivalent to that of a junior CS student, is preferred.

This course is aimed at junior- and senior-level Computer Science majors, but should be accessible to any student of life with some degree of mathematical and statistical maturity, reasonable experience with programming, and an interest in the topic area. If in doubt, e-mail me: fardina@umd.edu!

Tentative Topics:

Introduction to Data Science, Experiment Design, Introduction to Python, Data Types, Data collection, Git, Pandas, Database, SQL, Probability, Summary Statistics, Hypothesis Testing, Data Visualization, Data Exploration, Introduction to Machine Learning, Classifications, Decision Trees, Regression, Feature Engineering, ML Evaluation, Neural Network, Image Processing, Natural Language Processing (NLP) Introduction to Graph(s) Theory, Recommender Systems, Data Science Ethics

Learning Outcomes:

At the completion of this course, students will be able to:

- Develop a foundational understanding of data science, encompassing key concepts, methodologies, and tools.
- Proficiently use Python for data science applications, including data types, collection, and manipulation using libraries like Pandas.
- Explore machine learning concepts, including classifications, decision trees, regression, and feature engineering.
- Evaluate machine learning models using appropriate metrics, ensuring a comprehensive understanding of model performance.
- Understand advanced topics such as neural networks, image processing, natural language processing (NLP) etc.
- Understand the importance of applying ethical frameworks and guidelines to make informed decisions in real-world data scenarios.

By the end of the course, students will have acquired a versatile skill set, enabling them to tackle real-world data science challenges with confidence and ethical awareness.

Grading

There will be six assignments, one final project, two written midterm exams, and one final written exams (cumulative).

Final grades will be calculated as:

Component	Percentages
Assignments/Mini Projects	40%
Mid Exam 01	15%
Mid Exam 02	15%
Final Project/Tutorial	15%
Final Exam	15%

Late Policy

Late work gets no credit.

It is recommended to submit homework and projects on time. There will be a **15% penalty for late submissions of homework** and a **20% penalty for late submissions of project/tutorial checkpoints 1 and 2** within 24 hours after the deadline. **After this 24-hour period, no submissions will be accepted**. In **ELMS/Gradescope (as instructed in the assignment or project)**, you can submit multiple times, and only the last submission will be graded. The penalty for late homework will be applied automatically; no request is necessary. This policy applies to both homework and projects, EXCEPT for the **final submission/checkpoint 3 of the final project/tutorial and for any kind of BONUS work**

See the next section about how to contact us in special circumstances. We aim to help everyone succeed.

Communication & Logistics Request

We are going to use a combination of in-person and online office hours and the Piazza forum (
sign up here
) for Q&A. This means that it's appropriate to use Piazza for asynchronous communication with the course instructors and other students, and also for short, high-bandwidth discussions that could usually take place before/after class.

Note that Piazza is not appropriate for things like asking for accommodations, extensions or other such issues/concerns. For any logistics-related help, such as extensions or grading issues, please **fill out this form**: CMSC320 Logistics Request Form.
CMSC320- Logistic Request Form
However, If you do not receive a response or if your issue is not resolved within 48 hours, then please email:

- cmsc320gradingissues.dr.fardina@gmail.com
  for any grading, logistics, and extension issues

While sending an email, at the beginning of your email subject line, include relevant tags, such as `[hw_extension/req]`, `[project_extension/req]`, `[gradeissue]`, etc. More tags will be updated on the website.
If you have a request that fits into one of these categories and you *don't* email the address given above, then your request may not reach the right person and may not be answered in a timely manner.

If your correspondence does not fit into either of those two categories, please email an instructor (professor) **with `[CMSC320]` in the email subject line.** You may also go to an instructor's office hours at the times listed below.

Please note: if you don't include `[CMSC320]` in your subject line when emailing instructors, your email may not be filtered correctly.

Office Hours

** All hours are EDT*

Instructor Office Hours and TA office hours details can be found on ELMS as well as on Piazza.

Do keep an eye on Piazza, though; TAs will sometimes swap hours, shift hours, host hours on Zoom, and so on! Additionally, we have at least one TA explicitly covering Piazza on each weekday; all course staff will float around Piazza in general, too!

THERE ARE NO OFFICE HOURS FIRST WEEK (Sep 2 - Sep 5)

Schedule

ALL THE LECTURE SLIDES WILL BE POSTED HERE

The schedule is subject to change as the semester progresses!

#	Date	Topic	Reading	Slides	Notes
1	Sep 2, Tu	Introduction to Data Science and Data Type	Self-Study Slide: Git Additional Reading/Helpful slide: Python Demo Python	Lec1.1Intro Lec1.2DataTypes (Self-Study)	Sign up on Piazza !
2	Sep 4, Th	Experiment Design	Experimentation	Lec02ExpDesign	Final Project/Tutorial Instruction Out (Sep 3-4)
3	Sep 9, Tu	Pandas		Lec03Pandas	HW 1 Out (~~Sep 9~~Sep 10)
4	Sep 11 Th	SQL		Lec04_SQL	
5	Sep 16, Tu	Probability, Distributions, and Summary Stats	Probability (Bayes thm,Law of tot prob)	Lecture05_Probabilities_Part01	
6	Sep 18, Th	Probability cont.	Central Limit Thm Shared Description Material	Lec06_Probability_Part2	HW 1 Due (~~Sep 18~~Sep 19) HW 2 Out (Sep 18)
7	Sep 23, Tu	Hypothesis Testing	Hypo Testing Steps And Examples P-Value-Explaination	Lec07Hypothesis_Part01 Lec07Hypothesis_Part02 Lec07Hypothesis_Part03	
8	Sep 25, Th	Data Visualization		Lec08DataVisualization	Project First Checkpoint Due (Sep 25)
9	Sep 30	Data Exploration		Lec09DataExploration	
10	Oct 2, Th	Data Cleaning		Lec10DataCleaning_Part01	HW 2 Due (Oct 2) HW 3 Out (Oct 2);
11	Oct 7, Tu	Data Cleaning	HandlingMissingData	Lec11DataCleaning_Part02	
12	Oct 9, Th	Introduction to Machine Learning		Lec12_IntroToML	
13	**Oct 14, Tu**	**Fall Break (No class)**			
14	**Oct 16, Th**	**Midterm I**			HW 4 Out (Oct 16)
15	Oct 21, Tu	Feature Engineering & ML Evaluation		Lec13_FeatureEngineering	HW 3 Due (Oct 21)
16	Oct 23, Th	Feature Engineering & ML Evaluation cont.	Confusion Matrix CrossValidationVideo KFoldVideo	Lec14_ML_Evaluation	
17	Oct 28, Tu	Decision Tree	DecisionTree-Calculate-Entropy-InfoGain	Lec15_DecisionTree	Project Second Checkpoint Due (Oct 28)
18	Oct 30, Th	Classifications	Reading materials are given at end of the Classification lecture slide	Lecc16_Classification	
19	Nov 04, Tu	Regression	Simple Linear Regression	Lec16Regression	HW 5 Out (Nov 04)
20	Nov 06, Th	Unsupervised Learning & Dimensionality Reduction	PCA_short_slides	Lec17.1_DimReduction Lec17.2_UnsupervisedLearning	HW 4 Due (Nov 06)
21	Nov 11, Th	Introduction to Neural Network		Lec18.1_IntroToNeuralNetwork LEc18.2_NN_Training	
22	Nov 13, Th	Image Processing with CNN		Lec19CNN_ImageProcessing	
23	Nov 18, Tu	CNN continue and Intro to Natural Language Processing (NLP)		Lec20IntroToNLP	
24	Nov 20, Th	Intro to Natural Language Processing (NLP) cont.			HW 5 Due (Nov 20) HW6 Out (Nov 20)
25	**Nov 25, Tu**	**Midterm II**			
26	**Nov 27, Th**	**Thanksgiving Recess (No class)**			
27	Dec 2, Tu	Introduction to Graph Theory	NetworkX Intro to GraphQL These two materials are given for self-exploration if someone is interested.	Lec21_GraphTheory	
28	Dec 4, Th	Introduction to Graph Theory cont.	ClassNote_Girvan-NewmanAlgorithm Girvan & Newman. "Community structure in social and biological networks," PNAS-02.	Lec22_GraphTheory_Lecture2	HW 6 Due (Dec 4)
29	Dec 9, Tu	Recommender System		Lec23_RecommendationSystem_Lecture01	Project Final Checkpoint Due (December 9)
30	Dec 11, Th	Recommender System cont. and Data Ethics (self reading)		Lec24_RecommendationSystem_Lecture02 -- Lec25DataEthics (SELF STUDY)	
32	Final Exam Week	**Final Exam (Details)**		**Saturday December 20, 1:30pm - 3:30pm Reference: https://registrar.umd.edu/registration/register-classes/final-exams/fall**	

Deliverables

Assignments

All the Assignments will be **posted on Piazza/ ELMS.** Instructions will appear over the course of the semester. Most assignments get released on one or two days before the lecture material gets presented and are due one or two weeks after that.

#	Description	Date Released	Date Due
Homework 1	Git, Pandas, and SQL	Sep 9	Sep ~~18~~19
Homework 2	Python & Statistics & Hypothesis Testing	Sep 18	Oct 2
Homework 3	Data Exploration & Data Cleaning & Missing Data	Oct 2	Oct 21
Homework 4	Machine Learning: Classification and Clustering	Oct 16	Nov 6
Homework 5	Regression, Gradient Descent and Neural Network	Nov 4	Nov ~~20~~25
Homework 6	NLP/GraphTheory/Recommendation System/CNN (any mix)	Nov 20	Dec ~~04~~ 08

Final Project/Tutorial

There will be a group final project/tutorial with a maximum of four to six (four is recommended) persons in each group. Keep in mind that this is a semester-long project/tutorial for this course, and you should strive to make it your best work possible. It will be graded to a higher standard than the rest of the homework, considering that you have had the chance to practice these skills beforehand. **Project details will be posted on Piazza/ELMS. It will be your responsibility to decide project topic as well as project partners.**

#	Description	Date Released	Date Due
Checkpoint 1	Group Formation and Choosing Dataset	Sep 4	Sep 25
Checkpoint 2	Data preprocessing and Exploration	Sep 4	Oct 28
Final Checkpoint	Final deliverable of DS Project	Sep 4	Dec 9(`NO EXTENSION/LATE SUBMISSION IS ALLOWED)`)

Academic Integrity

Note that academic dishonesty includes not only cheating, fabrication, and plagiarism but also includes helping other students commit acts of academic dishonesty by allowing them to obtain copies of your work. You are allowed to use the Web for reference purposes, but you may not copy code from any website or any other source. In short, all submitted work must be your own. Cases of academic dishonesty will be pursued to the fullest extent possible as stipulated by the Office of Student Conduct. Without exception, every case of suspected academic dishonesty will be referred to the Office. If the student is found to be responsible for academic dishonesty, the typical sanction results in a special grade "XF", indicating that the course was failed due to academic dishonesty. More serious instances can result in expulsion from the university. If you have any doubt as to whether an act of yours might constitute academic dishonesty, please contact your TA or the course coordinator. The University of Maryland, College Park has a nationally recognized Code of Academic Integrity, administered by the Student Honor Council.B This code sets standards for academic integrity at Maryland for all undergraduate and graduate students. As a student, you are responsible for upholding these standards for this course. It is very important for you to be aware of the consequences of cheating, fabrication, facilitation, and plagiarism. For more information on the Code of Academic Integrity or the Student Honor Council, please visit http://www.shc.umd.edu.

Examples of Academic Integrity Violations:

The following are examples of academic integrity violations:

- Hardcoding of results in a project assignment. Hardcoding refers to attempting to make a program appear as if it works correctly (e.g., printing expected results for a test).
- Using any code available on the internet/web or any other source.
- Hiring any online service to complete an assignment for you.
- **You may not post the implementation of your assignments, materials related to the class (e.g., project description), or any other material associated with this online course except the final project. Even if the class is over and you have graduated, you may NOT post any material other than what is allowed.**
- Sharing your homework solutions or your "test code" with any student.
- Looking at or debugging another student's code.
- Using online forums to ask for help regarding our assignments.

AI Tools Policy for this class:

The use of AI tools (e.g., ChatGPT, Copilot, DALL·E) is strictly prohibited for any part of assignments, homework, quizzes, exams, or projects—including brainstorming, writing, coding, and editing. Exception: You may use AI tools only for brainstorming ideas for the semester-long final project. If you do, you must clearly declare this use in your project checkpoints submission and cite the tool as a reference. AI-generated content or code cannot appear in your final deliverable. This restriction also applies when using Google Colab or any other online coding platform—AI-assisted features must be turned off. Violating this policy will result in a score of zero on the assignment/project and may lead to an academic integrity referral. While AI tools will play a role in your future work, this policy is designed to help you develop original ideas and a unique voice—both essential skills for this course. If you have questions or would like to suggest potential exceptions, please email me at [instructor email], and I will be happy to discuss further
Consequences for non-compliance: Failure to adhere to this policy may result in a zero on the particular course work where the AI tool is used. In addition, the university honor code is applicable here: violation of the honor code and appropriate action will be enforced.

Class Announcements

You are responsible for reading the class announcements that are posted on both the course webpage and ELMS. Please check them often (at least once a day). Important information about the course (e.g., deadlines, assignment updates, etc.) will be posted on the course webpage.

Additional Administrative Information

University Policies and Resources

Policies relevant to Undergraduate Courses are found here:
http://ugst.umd.edu/courserelatedpolicies.html
. Topics that are addressed in these various policies include academic integrity, student and instructor conduct, accessibility and accommodations, attendance and excused absences, grades and appeals, copyright and intellectual property.

Projects/Labs: On any graded project or lab, you are NOT allowed to exchange code. We compare each student's code with every other student's code to check for similarities. Every semester, we catch an embarrassingly high number of students that engage in cheating and we have to take them to the Honor Council.

GroupMe/Other Group Chats: We encourage students to talk about course material and help each other out in group chats. However, this does NOT include graded assignments. There have been a couple instances in the past where students have posted pictures/source files of their code, or earlier sections have given away exam questions to later sections. Not only did this lower the curve for the earlier section because the later one will do better, the WHOLE group chat had to pay a visit to the Honor Council. It was an extremely ugly business. Remember that in a group of 200+, someone or the other will blow the whistle. If you happen to be an innocent person in an innocent groupchat and someone starts cheating out of the blue, leave it immediately (and better yet, say you are leaving and say you will report it).

Github: You may post your project code to private Github (or similar service) repos only. As a student, you can make a private repo for free. Just remember that your free premium subscription has an expiration date, and your code becomes public once it expires. The Honor Council can retroactively give an XF (even to students who have already graduated) if your code is then used by another student to cheat. So just be careful. Posting graded code to a public repo will give you a free ticket to the Honor Council(Unless the instructor has given you permission with some strict conditions).

Excused Absences

Missing an exam for reasons such as illness, religious observance, participation in required university activities, or family or personal emergency (such as a serious automobile accident or close relative's funeral) will be excused so long as the absence is requested in writing at least 2 days in advance and the student includes documentation that shows the absence qualifies as excused; a self-signed note is not sufficient as exams are Major Scheduled Grading Events. For this class, such events are the final project assessment and midterms, which will be due on the dates listed in the schedule above. The final exam is scheduled according to the University Registrar.

For medical absences, you must furnish documentation from the health care professional who treated you. This documentation must verify dates of treatment and indicate the timeframe that the student was unable to meet academic responsibilities. In addition, it must contain the name and phone number of the medical service provider to be used if verification is needed. No diagnostic information will ever be requested. Note that simply being seen by a health care professional does not constitute an excused absence; it must be clear that you were unable to perform your academic duties.

It is the University's policy to provide accommodations for students with religious observances conflicting with exams, but it is the your responsibility to inform the instructor in advance of intended religious observances. If you have a conflict with a planned exam, you must inform the instructor prior to the end of the first two weeks of the class.

The policies for excused absences do not apply to project assignments. Projects will be assigned with sufficient time to be completed by students who have a reasonable understanding of the necessary material and begin promptly. In cases of extremely serious documented illness of lengthy duration or other protracted, severe emergency situations, the instructor may consider extensions on project assignments, depending upon the specific circumstances.

Besides the policies in this syllabus, the University's policies apply during the semester. Various policies that may be relevant appear in the
Undergraduate Catalog
.

If you experience difficulty during the semester keeping up with the academic demands of your courses, you may consider contacting the Learning Assistance Service in 2201 Shoemaker Building at (301) 314-7693. Their educational counselors can help with time management issues, reading, note-taking, and exam preparation skills.

Information regarding official University closing and delays

Students can find information about official university closings and delays on the campus website or by contacting the weather emergency phone line at 301-405-7669. If any exam or class assignment is rescheduled or a class is canceled due to inclement weather, we will provide announcements via ELMS. We will ensure to follow the university campus rules

Right to Change Information

Although every effort has been made to be complete and accurate, unforeseen circumstances arising during the semester could require the adjustment of any material given here. Consequently, given due notice to students, the instructors reserve the right to change any information on this syllabus or in other course materials. Such changes will be announced and prominently displayed.

Course Evaluations

Course evaluations are important and the department and faculty take student feedback seriously. Near the end of the semester, students can go to
http://www.courseevalum.umd.edu
to complete their evaluations.

Miscellaneous Resources

As a textbook reference, use this book
CMSC320TextBook
As we go through the course sometimes I will mention additional resources or next steps. None of this is required for the course, but students have asked for me to keep a record of which texts/websites I mention.

- Python for Data Analysis
  covers some of the same topics that we cover in this class, but in textbook form.
- Data Science From Scratch
  this book is meant to give you a grounding in how some of the statistics and math could be implemented. Mostly, data scientists use off-the-shelf libraries for their mathematical routines/functionality, this books works through how you would implement some of those libraries. The idea is that this might give you a deeper understanding of what is going on. The implementations in this book are not meant to be high-performance, or industrial strength, but illustrative.
- Web Scraping 101
  is a good overview to how web scraping works.

COMPUTER SCIENCE
UNIVERSITY OF MARYLAND

Previous versions of this course:
Fall 2021
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Summer 2021
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Spring 2021
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Fall 2020
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Fall 2019
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Fall 2018
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Fall 2017
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Spring 2017

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Web Accessibility